

PHYSICS

CONCISE NOTES

For Senior High Schools

- ✓ Clear Explanations
- ✓ Worked Examples
- ✓ Exam-Focused Questions
- ✓ Diagrams & Applications

$$E = mc^2$$


WAVES
& OPTICS


ELECTRICITY &
MAGNETISM


HEAT &
THERMAL PHYSICS


ATMOSPHERIC &
MODERN PHYSICS

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Dedication

This book is dedicated first to the loving memory of my late father, Simeon Nyatuka, whose life, guidance and sacrifices remain a lasting source of inspiration.

To my mother, Hyrine Kerubo, for her love, strength, encouragement and unwavering support.

To my guardians, Peter Munde Mong'are and Teresa Kwamboka Sawe, for their guidance and support throughout my journey.

To my friends Mnaree, Richie, Duke, Caleb, Cuba, Philo, Rob, Tim, Mr Peter Kimani, and the many others whose encouragement, friendship and support have contributed to my growth.

To Her, for the inspiration, encouragement and memories that have been part of the journey.

To my fellow teachers, educators and mentors who continue to shape young minds.

And most importantly, to every learner who opens this book with a desire to understand Physics, improve and achieve more.

May your curiosity continue to grow, and may every difficult problem become another opportunity to discover what you are capable of.

— Hezekiah Omweri Nyatuka

Preface

Physics is not simply a collection of equations. It is a way of understanding motion, energy, light, electricity, the environment, technology and even our exploration of space. This book has been developed to help Grade 11 learners build clear Physics concepts while connecting classroom learning with practical situations and problem solving.

CONCEPT → UNDERSTAND → OBSERVE → CALCULATE → APPLY → PRACTISE → THINK BIG

Learners are encouraged not merely to memorise formulas, but to understand why physical phenomena occur, how quantities are related and how Physics can be applied to real situations.

How to Use This Book

REMEMBER: Important facts and relationships.

HINT: Guidance for solving problems.

NOTE / COMMON ERROR: Mistakes and misconceptions to avoid.

WORKED EXAMPLE: Step-by-step calculations.

PRACTICE: Immediate learner practice.

PRACTICAL / INQUIRY: Investigation and observation.

APPLICATION: Physics in everyday life.

THINK BIG: Reasoning and deeper thinking.

EXAM FOCUS: Assessment preparation.

PEAK ACHIEVERS CHECKPOINT: End-of-topic revision.

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PART I

CONCISE NOTES

STRAND 1: MECHANICS AND THERMAL PHYSICS

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CHAPTER 1 — FLUID FLOW

A fluid is a substance that can flow. Liquids and gases are fluids.

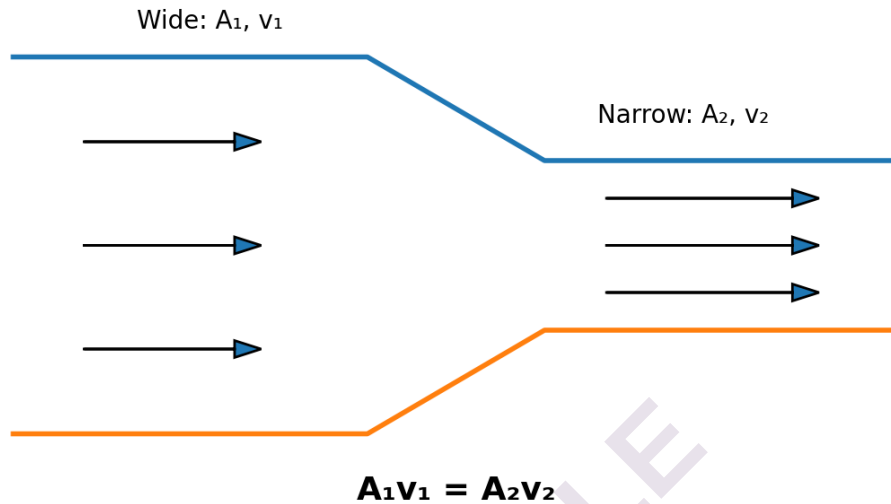


Figure 1.1: Continuity in a pipe of changing cross-sectional area.

1.1 Types of Flow

In streamline (laminar) flow, fluid particles follow smooth paths or streamlines. In turbulent flow, the motion is irregular and mixing occurs.

REMEMBER: Streamline flow is orderly; turbulent flow is irregular.

1.2 Volume Flow Rate

$$Q = V/t$$

For a fluid moving through a pipe of cross-sectional area A with speed v :

$$Q = Av$$

1.3 Equation of Continuity

For steady flow of an incompressible fluid, the volume flow rate is the same at every cross-section.

$$A_1v_1 = A_2v_2$$

REMEMBER: If cross-sectional area decreases, fluid speed increases for steady incompressible flow.

Worked Example

Water flows through a pipe of area 6 cm^2 at 2 m/s and enters a section of area 3 cm^2 . Find the new speed.

$$6 \times 2 = 3 \times v_2$$

$$v_2 = 4 \text{ m/s}$$

1.4 Bernoulli Principle

For suitable steady ideal flow, an increase in fluid speed can be associated with a decrease in static pressure. Applications include atomisers, Venturi arrangements and ventilation.

1.5 Viscosity

Viscosity describes a fluid's resistance to flow. It depends on the nature of the fluid and temperature.

APPLICATION: Fluid-flow ideas are applied in irrigation, water supply, drainage, ventilation, medicine and industry.

PEAK ACHIEVERS CHECKPOINT

1. Define a fluid.
2. Distinguish streamline flow from turbulent flow.
3. State the equation for volume flow rate.
4. State the equation of continuity.
5. A pipe narrows from 8 cm^2 to 4 cm^2 . If water moves at 2 m/s in the wider section, calculate the speed in the narrower section.
6. State the qualitative Bernoulli relationship between speed and static pressure under suitable conditions.
7. Give two applications of fluid-flow principles.
8. What is viscosity?

CHAPTER 2 — LINEAR MOTION

Linear motion is motion along a straight line. It is described using distance, displacement, speed, velocity and acceleration.

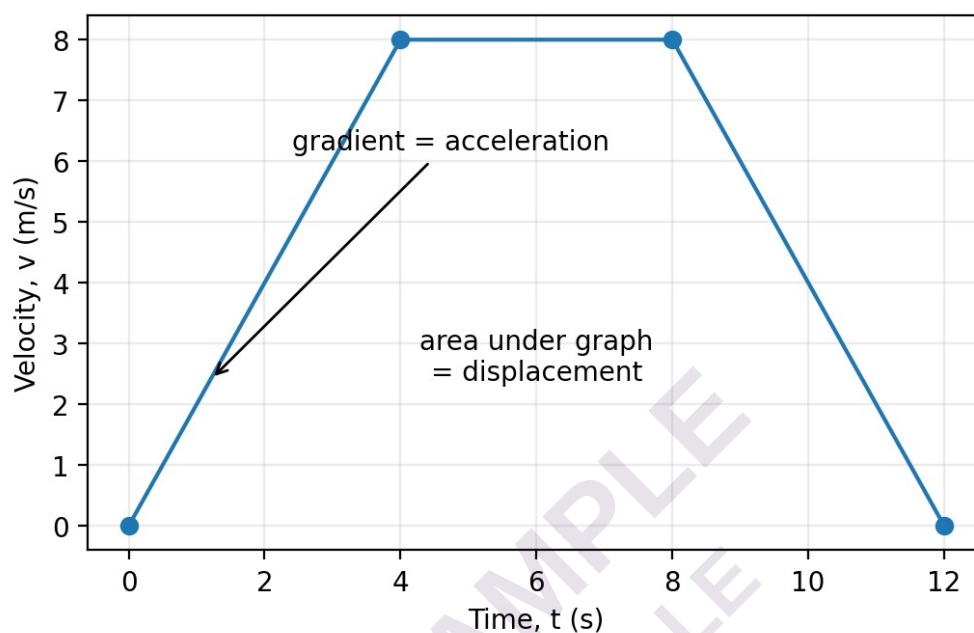


Figure 2.2: Velocity-time graph: gradient gives acceleration and area gives displacement.

2.1 Distance and Displacement

Distance is the total path length travelled. Displacement is the straight-line change in position in a specified direction.

2.2 Speed and Velocity

$$\text{speed} = \text{distance}/\text{time}$$

$$\text{velocity} = \text{displacement}/\text{time}$$

2.3 Acceleration

$$a = (v - u)/t$$

COMMON ERROR: Negative acceleration does not always mean an object is slowing down; the result depends on the directions of velocity and acceleration.

2.4 Uniformly Accelerated Motion

For constant acceleration:

$$v = u + at$$

$$s = ut + \frac{1}{2}at^2$$

$$v^2 = u^2 + 2as$$

$$s = \frac{1}{2}(u + v)t$$

Worked Example

A vehicle increases its speed from 10 m/s to 25 m/s in 5 s.

$$a = (25 - 10)/5 = 3 \text{ m/s}^2$$

2.5 Motion Graphs

Gradient of a displacement-time graph gives velocity. Gradient of a velocity-time graph gives acceleration. Area under a velocity-time graph gives displacement.

2.6 Free Fall

Near Earth's surface, $g \approx 9.8 \text{ m/s}^2$. Use 10 m/s^2 when a question states or permits that approximation.

2.7 Road Safety

Stopping distance is affected by reaction time, speed, road conditions, tyres and brakes.

PEAK ACHIEVERS CHECKPOINT

9. Distinguish distance from displacement.
10. Distinguish speed from velocity.
11. Define acceleration.
12. State the four equations of uniformly accelerated motion used in this chapter.
13. What does the gradient of a velocity-time graph represent?
14. What does the area under a velocity-time graph represent?
15. A body starts from rest and accelerates at 2 m/s^2 for 6 s. Find its final velocity.
16. State two factors that affect stopping distance.
17. State the approximate value of g near Earth's surface.

CHAPTER 3 — PROJECTILE MOTION

A projectile is an object that, after being launched, moves mainly under the influence of gravity when air resistance is neglected.

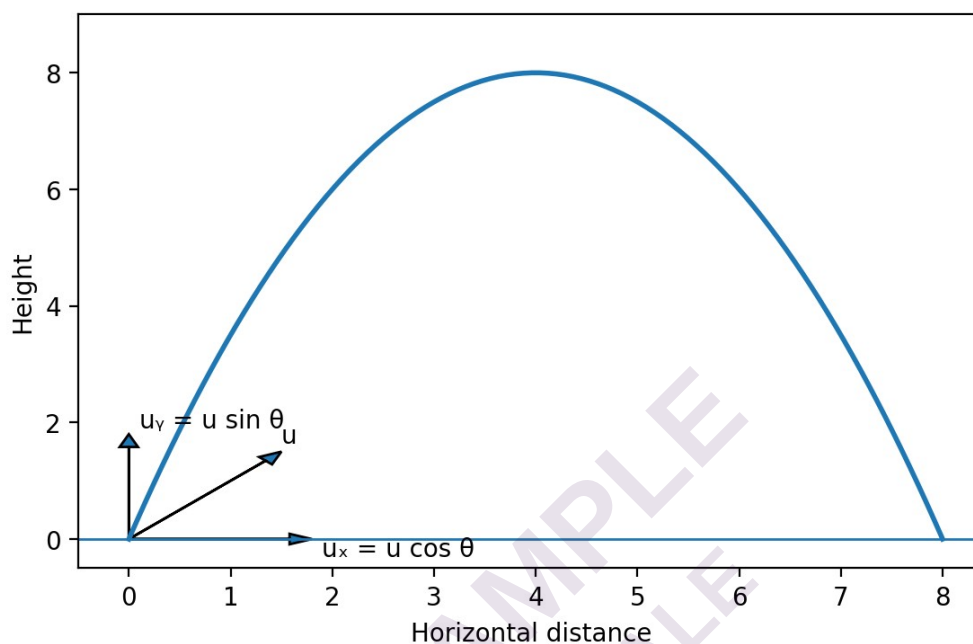


Figure 3.1: Resolving projectile velocity into horizontal and vertical components.

3.1 Velocity Components

$$u_x = u \cos \theta$$

$$u_y = u \sin \theta$$

In ideal projectile motion, horizontal acceleration is zero and vertical acceleration is downward with magnitude g .

$$x = (u \cos \theta)t$$

3.2 Maximum Height

$$H = u^2 \sin^2 \theta / 2g$$

3.3 Time of Flight

For launch and landing at the same vertical level:

$$T = 2u \sin \theta / g$$

3.4 Horizontal Range

For launch and landing at the same vertical level:

$$R = u^2 \sin 2\theta / g$$

REMEMBER: The 45° maximum-range result applies to ideal same-level projection with negligible air resistance.

3.5 Horizontal Projection

$$t = \sqrt{(2h/g)}$$

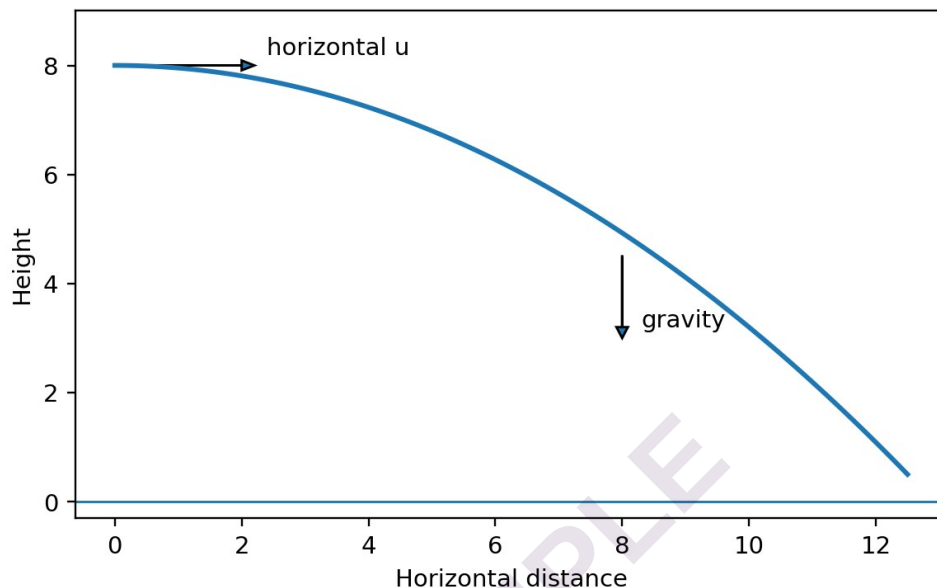


Figure 3.2: Horizontal projection combines constant horizontal motion with vertical acceleration due to gravity.

$$x = ut$$

Worked Example

A ball is projected horizontally at 8 m/s from a height of 20 m. Take $g = 10 \text{ m/s}^2$.

$$t = \sqrt{(40/10)} = 2 \text{ s}$$

$$\text{range} = 8 \times 2 = 16 \text{ m}$$

THINK BIG: Why are the horizontal and vertical components of ideal projectile motion treated independently?

PEAK ACHIEVERS CHECKPOINT

18. Define a projectile.
19. Write expressions for the horizontal and vertical components of initial velocity.
20. State the horizontal acceleration of an ideal projectile.
21. State the vertical acceleration of an ideal projectile.
22. State the formula for maximum height.
23. Under what condition can $T = 2u \sin\theta/g$ be used?
24. Under what condition can $R = u^2 \sin 2\theta/g$ be used?
25. Why is 45° special in ideal same-level projection?
26. A ball is projected horizontally at 12 m/s from a height of 20 m. Find its time of flight and horizontal range using $g = 10 \text{ m/s}^2$.

PEAK ACHIEVERS

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